

Claude Code — a working day

A study guide for an agentic coding tool: what it changes about the job, what it does not, and where the leverage actually is.

Each section states an idea, makes the argument for it, and points at the lab that tests it. The labs live in [cc-exercises/](#); each has its own brief.

On sources. This material is original. It is informed by Ken Kousen's *Claude Code: Up and Running* (O'Reilly), and none of its text or prompts is reproduced here — where a move is his, the section cites the chapter and describes what the prompt has to accomplish so you can write your own. The experiment in §4, the fixtures, and the verifier are this repo's.

0. Three scales, and why the middle one matters

Lab 00. Two projects of the same size on different scales. Measure both, then find what the measurements missed.

A useful way to size work before you start:

- a **doghouse** — one file, no tests, no users but you. Mistakes cost minutes.
- a **cabin** — a few modules, real tests, someone else will read it. Mistakes cost an afternoon.
- a **skyscraper** — a system with history, dependencies and downstream consumers. Mistakes cost a quarter.

Kousen opens with this framing (ch. 1) and it is worth stealing, because the interesting thing is what it predicts: an agentic tool is spectacular at doghouses, genuinely useful at cabins, and needs the most supervision exactly where the stakes are highest. Your working life is mostly cabins and skyscrapers. **A demo is a doghouse. Judge the tool on the cabin.**

That is also why labs 01 through 07 build one small app and then keep going with it. The first hour is the doghouse; everything after is what happens when the same code has to survive a second week.

***The model is a knob.** Speed and depth are a setting, not an identity — reach for a faster model on mechanical work and a deeper one on design. Most people leave the knob where it was and then form an opinion about "the tool".*

1. First contact — the part that is genuinely new

Lab 01. Build something that runs, from nothing, in one session.

What is new is not code completion. It is that the tool reads your files, decides which ones matter, edits several of them, runs the result and reads the error. The loop closes without you in it.

Two habits to form in the first hour:

Ask for a read-only orientation before you ask for a change. "Tell me what this does, change nothing" costs one turn and has zero blast radius. It also tells you whether the thing in front of you understood the repo before it starts rewriting it. (Ch. 2 has his version of that prompt.)

Say how big the first version should be. Left unconstrained, a request for "a small app to practise memorising a poem" comes back with argument parsing, a config file and a package layout. Name the scale: *one file, standard library only, no packaging, no argument parsing*. You can always grow it.

And decline the offer at the end. It will propose next steps. Most of them are real. Accepting all of them is how a doghouse becomes an unmaintained cabin in forty minutes.

2. Getting what you asked for

Lab 02. One failing test, one prompt, and a defect that appears twice.

The difference between a vague request and a precise one is not politeness. It is that a vague request leaves the decisions to the model, and you will not be told which decisions those were.

Treat a prompt as a message to a colleague about to do the work: enough context to decide well, a picture of what success looks like, the constraints they have to respect. That framing also says what to leave out — you would not tell a colleague which variable names to use.

Five jobs, and a good prompt does all five in about six lines:

Job	Because
The symptom , in checkable terms	Not "validation is broken" — a status code you can assert on
The desired behaviour , as its own sentence	A separate claim from the symptom. Both are needed
The likely location	A real path. Saves a search, and a search is where the tokens go
The local convention , by pointing at it	Cheaper and more exact than describing it

Job	Because
The boundary	One line, and it is the difference between a reviewable diff and a surprising one

The longer prompt is also the **cheaper** one to run: a vague request sends the agent through rounds of discovery to work out what you meant, and those rounds cost more than the sentence you did not write. Kousen's ch. 3 runs this as a before/after pair — a two-word request against a six-line one. Both are his; the lab asks you to write yours.

The sentence that carries the module

The same constraint, three ways:

```
Don't put validation in the handler
Put validation in a validator function
Put validation in a validator function so every path that accepts user
input is checked the same way
```

The first can be obeyed while completely missing the point. The second names a target. The third is the one to write, because **the agent generalises from the reason** — it makes consistent choices in places your prompt never mentioned.

A reason is a rule that transfers. An instruction is a rule that does not.

Lab 02 measures that instead of asserting it. The app has the same defect in two handlers; the failing test covers one, and nothing anywhere mentions the other. A prompt that names the handler fixes one. A prompt that names *why* tends to fix both — and the checker reports which one you wrote.

Two more things the lab measures, because they fail independently:

- **Literalism.** Recent models take instructions literally and do not carry one case to the next. Say *empty* and you get exactly empty; a whitespace-only title still walks through. Scope is something you state, not something you hope it infers.
- **The fence.** The fixture ships a genuinely ugly module with a `TODO` on it that nothing asked you to fix, and a test file that a red-to-green shortcut would love to edit. Both are frozen and checked by hash. A boundary clause that is too vague to hold is one you find out about here rather than in review.

Ask before you request

Two verbs, and most people request too early.

Verb	Goal
Ask	"How does this app validate input today?" — understanding

Verb	Goal
Request	"Add title validation to both handlers." — action

Keep them in separate prompts. Ask it to explain and change in one shot and you get a confident edit built on an explanation you never checked — and if the explanation was wrong, so is the change.

Then push back on one decision without asking for a change. *"Why did the check go in the handler rather than the validator?"* is a question, and you get a defence or a concession — either is information. Phrased as an instruction, you get an edit and learn nothing. People mix the two constantly and then complain the tool is too agreeable.

3. Driving the session

Lab 03. Measure the window against an estimate, then pollute it on purpose and watch the answers degrade with space to spare.

Context is the budget you are actually spending. Every file read, every tool result, every one of your own turns stays in it, and turn fifty carries turns one through forty-nine on its back. Two consequences:

Long sessions get worse, not better. Not because the model tires, but because the signal-to-noise ratio of what it is reading drops. Three abandoned approaches and a stack trace from an unrelated bug are still in there competing with your actual question.

Clearing is a tool, not a failure. Finish a piece of work, write down what survived — in `CLAUDE.md`, in a commit message, in a scratch file — and clear. The written artifact is what carries forward; the conversation is not.

Compaction happens automatically when you run out of room, and it is lossy in a specific way: **contracts survive compaction, procedures fray.** A summary keeps "storage is injected, tests never touch the disk" and loses "then we tried three approaches and the second one failed because of X". So write down constraints, not narratives. If something must survive, give it to a file.

Modes. Plan mode proposes and does not edit. Accept-edits stops asking about each write. Auto is the default on paid plans and is not the dangerous one people assume — the boundary is set by your permissions, not by the mode. Know which you are in. The failure is not "the wrong mode", it is not knowing.

Ask for a plan in words before a plan in files. A plan you can read in fifteen seconds is a plan you can reject in fifteen seconds. Then one question, worth memorising: *"what are the alternatives, and why is this one better?"* It is the cheapest defence against a confident plan that is merely the first plan.

4. Tests force the design — sometimes

Lab 04. A bug is already fixed and the suite is already green. Audit it.

Here is the claim, stated so you can disagree with it:

Ask an agentic tool for tests on a messy app and it will not just add tests — it will restructure the code so that tests are possible, and that restructuring is better design you got by asking for something else.

We ran it instead of asserting it. **Nine sessions, three starting points, one prompt that never mentions structure, modules or refactoring.** Read-only, so the tool could only propose.

Starting point	Restructured	Declined
JavaScript — logic inside an inline <code><script></code>	2 of 3	1
Python CLI — everything inside <code>main()</code>	0 of 3	3
Python Flask — <code>requests.get</code> inline in the route handler	0 of 3	3

The claim is language-dependent, and in Python it is false.

The reason is structural, not a model whim. An inline `<script>` exposes nothing — there is nothing to import, so the only path to a test is to make something importable, and the design has to move. **A Python module is always importable.** Globals can be reassigned, `input` can be faked, stdout can be captured, `monkeypatch` is right there. The cheapest path to a test never requires touching the design, and every run found it and reasoned it out loud.

So: **in Python, the tooling will not push back on you.** If you want the seam, ask for it. Three things follow.

1. It will not run the tests unless you tell it to. Do not assume the loop closes itself.

2. "The smallest change" is not a magic word. Both JavaScript runs we kept transcripts for chose Node's built-in test runner — a perfectly reasonable reading of "smallest", and not the structure the lesson needed. Run it twice, get two structures. If your repo needs a particular shape, name the shape.

3. A red test goes green two ways. Fix the code, or quietly weaken the test. Both satisfy "until the suite passes", and a model optimising for a green suite has no intrinsic preference between them. This is why *"it wrote 15 tests"* is a meaningless number: fifteen tests adjusted to fit the code are worse than none, because they are now load-bearing lies. Ask for two lists when it finishes — **what changed** and **what you verified** — and read the second one.

The sharpest finding was about invariants. The one sentence cited most often across the nine runs for *not* touching the design was the prompt's own invariant — *keep the behaviour*

unchanged. It is a reasonable sentence, it protects something real, and it blocked the very thing the prompt existed to demonstrate, six times out of nine. **Write your invariants knowing they will be obeyed literally**, and pair every one with a licence to object: *if this constraint makes the task awkward, say so instead of working around it*.

So stop treating a green suite as an answer

Put those three together and the conclusion is uncomfortable: **the suite is the weakest link in the chain, and it is the one everybody reads**. The chain that actually justifies "this is fixed" runs:

```
failure observed → root cause → minimal change → evidence → invariants → challenge
```

A green suite occupies one link. On its own it tells you that some assertions passed — not that they were the right assertions, not that they were the *same* assertions as yesterday, and not that nothing else moved.

Lab 04 is that audit, and you do not write the fix. You are handed a repository where an agent already fixed the bug: two commits, a green suite, an honest commit message, a three-line diff. Three things in it do not survive contact with evidence, and one of them is invisible from behaviour alone — the test written to catch the bug was widened to accept both answers, so it now passes whether the bug is present or not. The code happens to be correct today, which is exactly why nobody looks, and exactly why the next regression will ship.

The six moves, in order, and the order matters because five of them happen **before** you let it edit anything:

1. **Failure observed.** Does the original test actually fail on the pre-fix code? No red, no meaningful green.
2. **Root cause.** Ask for the code path, not a summary. A diagnosis you cannot trace in the file is a guess.
3. **The change.** Is it minimal? What *other* inputs does the new condition affect? Is there a validator that should have been reused?
4. **What else changed.** Read the whole diff. Production, tests, and everything else, separately. "*While I was in there*" is where unreviewed changes live.
5. **Confirm** — and for every claim, **show the supporting evidence**. That last clause is the whole discipline. You do not want conclusions; you want claims plus what backs them.
6. **Challenge.** Ask it to argue its own fix is wrong. What did it assume about the input? In the lab, this is the step that finds the crash.

The agent operates the tools. You audit the evidence. It can read the repo, form hypotheses, edit, run `pytest`, read `git diff`. Your job is not to repeat any of that by hand. It is to decide whether the chain of evidence actually justifies the claim.

5. Project memory

Lab 05. Write `CLAUDE.md` by hand, then have it reviewed. Scope one rule. Then put a third instruction somewhere that refuses, and measure which of the three you were actually obeyed on.

`CLAUDE.md` is prepended to every session in the project. That is the whole design, and both of its consequences follow from it.

Write it yourself first. `/init` generates a description of the code — which is the part the agent can already get by reading the code. The value is the part it cannot: *standard library only, only `cli.py` may construct real storage, if `cli.py` grows a conditional, the rule belongs in a module*. Draft those by hand, then ask what a new contributor would get wrong, then cut.

Keep it short. A 300-line memory file is 300 lines of context on every request, including the ones where none of it is relevant. Five sections, under a hundred lines: what this is, architecture, commands, conventions, workflow.

Then scope the rest. Advice about tests is noise while you are editing `storage.py`. A file under `.claude/rules/` with `paths: frontmatter` arrives only when a matching file does:

```
---
paths:
  - "tests/**/*.py"
---

# Testing rules

- Tests must not touch the filesystem, stdout, or a real clock.
```

That is the difference between memory and a rule: memory is always loaded; a rule is loaded when it is relevant. Most `CLAUDE.md` files that have gone bad are rules that were never scoped.

`@-imports` in `CLAUDE.md` resolve at launch, not lazily — importing a large document costs you its whole size on every session, so import small things.

6. Permissions and the trust boundary

Lab 06. Write a deny list for a Flask app with a live API key in `.env`.

Three ingredients together are what make an agent dangerous. Any two are fine:

1. access to **private data**,
2. exposure to **untrusted content**,
3. the ability to **communicate outward**.

Your editor session usually has all three: your repo, a dependency's README or a web page it fetched, and `curl` or `git push`. Deny rules are how you break the set.

Deny is the only list that is not a convenience. An allow list you get wrong costs you a prompt. A deny list you get wrong costs you a secret. So spend your attention there, and cover the variants — `Read(.env)` does not match `.env.local`, `.env.production` or `.env.staging`, which hold the same secrets.

```
{
  "permissions": {
    "deny": ["Read(.env)", "Read(.env.*)", "Bash(git push *)"],
    "allow": ["Bash(python3 -m pytest *)"]
  }
}
```

Note that `allowed-tools` in a skill or subagent **grants**; it does not restrict. The fence is `permissions`.

Say the boundary out loud as well. Settings are the fence; a sentence in the conversation — *don't touch anything under `migrations/`, and don't push* — is what stops it proposing work you will have to decline five times. Both, not either.

A prompt worth keeping: make it classify the work before it does the work. *Is this a one-line fix, a contained change, or something that touches the data model?* You get the blast radius before you get the diff, and you can stop at the answer. (Ch. 5 has his version, along with the same feature written at three levels of trust.)

7. Skills and hooks

Lab 07. Write a skill that reports coverage gaps, and a hook that formats Python. Then break the description on purpose and watch the routing stop — the description is the whole interface, and nothing else in the file is read until it matches.

The distinction decides which one you should be writing:

- a **skill** is a procedure the agent reaches for **when the situation matches**;
- a **hook** is code that runs **whether or not anyone reached for anything**.

"Run the formatter after every edit" is a hook. "Here is how we investigate a coverage gap" is a skill. Getting this backwards produces a skill nobody invokes, or a hook that fires on work it has no business touching.

The description is the entire interface

Only a skill's frontmatter is loaded at startup; the body is read after the skill has already been chosen. So the description has to carry both **what it does** and **when to use it**, phrased the way

someone would actually ask — *what is untested, where are the coverage holes, is this good enough to open a pull request.*

Two failures, both common:

- A description that says only what the skill does. It never fires, because nothing in it matches how the question gets asked.
- **An unquoted colon.** `description: Reports coverage: gaps and risks` is a nested mapping in YAML. Your description silently truncates at `Reports coverage` and the skill mysteriously stops matching. Quote it.

Test a skill by **not** naming it. If it does not fire on the natural question, the description is the bug, not the body.

Hooks fail in the other direction

A hook runs on every matching tool call, on every machine, including the one that does not have your linter installed. So:

- **exit 0 when the tool is missing.** A hook that errors because `ruff` is absent turns every edit into a failure for someone who never asked for the check.
- **PostToolUse cannot block an edit** — it fires after the write has happened. If you want to stop something, you need `PreToolUse` and exit code 2.
- Ask to see both files before they are written. You are installing code into your own loop.

8. Where the leverage actually is

Lab 08. Connect an MCP server, read what it costs, then do the same job with a command and a skill and read that.

By this point you have a tool that edits files, remembers your conventions, respects a fence and runs procedures. The instinct is to reach for more automation. The better question is where the time actually goes.

It is not typing. It was never typing. It goes into **deciding what to build, finding out how the existing thing works, and establishing that a change is correct.** An agentic tool helps enormously with the middle one, somewhat with the third, and not at all with the first — and the first is the one that determines whether the work mattered.

So the leverage is in the parts that are cheap to write and expensive to skip: the spec (§10), the review (§11), and the orientation (§12). Everything else is speed on work you had already decided to do.

Which is why the lab here is a protocol you probably do not need. MCP is the right answer when Claude has to reach a system it genuinely cannot otherwise reach — a service another team owns, a live database, an API with a schema that changes under you. It is the wrong

answer when a local command would have done, and the cost is not hypothetical: every tool definition is resident in the window for the whole session, paid for on every turn, used or not. The lab has you measure both and decide with the numbers in front of you. Most of the time the answer is the smaller thing, and knowing *why* is worth more than the reflex.

9. Subagents and parallel work

Lab 09. Review a change with a subagent you wrote, with `/context` read either side of both rounds — and triage what comes back, rejecting at least one finding.

A subagent is **a fresh context with a narrow job and a reporting contract**.

The fresh context is the point. A reviewer that watched you write the code has already been convinced by your reasoning — it read the argument for the approach before it read the approach. One that starts cold sees the diff the way a colleague does.

Four things a subagent file needs:

- **frontmatter** with a `name` and a `description` that says *when to delegate to it*. Same rule as a skill: this is what gets matched.
- **a tool list**. An unrestricted subagent is just another session. A reviewer gets read and search tools and nothing that writes. The narrowing is what makes it trustworthy enough to run unattended.
- **standards to review against**, not "review this code".
- **a reporting contract**: what comes back and in what shape. Without it you get whatever it felt like summarising, and you cannot act on a summary.

Invoke it through its interface. If you have to re-explain the job every time, the file is not earning its place.

On running several at once: read-only lenses parallelise freely — a coverage reviewer and a naming reviewer over the same diff cannot collide, and the second routinely sees what the first did not. Writers need **disjoint ownership and a stop condition**, or you get two agents editing the same file and one of them losing. (Ch. 8 has a full worked anti-example of exactly that.)

Two lenses is not two opinions. It is two coverage areas, and neither covers everything — so one step of the lab is saying what the review missed.

Three costs, and only the first is obvious.

- **Tokens**. The subagent opens the same files, and pays for a fresh system prompt and a fresh tool list on top. The total goes up, not down; what changes is *where* the tokens are spent. You moved a cost out of your window. You did not take it off the bill.
- **Coordination**. Someone has to write the brief. A thin brief comes back as a confident answer to a question you did not ask.

- **Verification.** The report arrives with no working shown, and every line of it is a claim until you check it. This is the cost that does not shrink when you delegate, and the one nobody budgets for.

Which is why the second half of lab 09 is a measurement rather than an argument. You survey a package you have never opened twice — once in the main session, once through the subagent you just wrote — with a `/context` reading on each side of both rounds. The four numbers subtract to a saving you write down even when it comes out at zero, which is the more interesting result and the one worth trusting, because nobody re-runs a measurement until it looks worse. Then you paste the report through `triage.py`, which splits it into numbered findings, and mark every one accept or reject with a reason under it. At least one of them is wrong: a reviewer that read the code and not the docstring beside it is confidently wrong in exactly the register that reads like being right. Catching that one is what verification costs, and paying it on purpose beats discovering the bill later.

10. From a prompt to a pull request

Lab 10. Cut an eight-bullet issue down to one slice, write the contract, and let a gate tell you which lines are still only sentences.

The move that makes this work is **narrowing first**. A backlog item is not a task; it is a heading. Before anything gets written, cut it to one slice with an obvious boundary — and get the alternatives named out loud so the cut is a decision rather than the first idea.

Then the spec. The thing that distinguishes a spec from a task description:

- **postconditions**, not steps. *After this change, an invalid saved index resolves to line 0* — not *add a validation function*. Steps are your design smuggled into the request; postconditions leave the design open and still let you check the result.
- **invariants**, stated as invariants. *Behaviour unchanged. No new dependency. Public API unchanged.*
- **an explicit out-of-scope list**. This is the one people skip and then relitigate in review.

Write invariants knowing §4's finding: they will be obeyed literally, including in the cases where obeying them defeats the purpose. So pair them with a door — *if a constraint makes this awkward, say so rather than working around it*.

The payoff is that when the result is wrong, you can point at the postcondition it violated instead of rewriting the task from scratch. That is the difference between a correction and a restart.

11. Review and recovery

Lab 11. Break something on purpose. See whether the review names the layer.

Review the verifier before you review the change. If the suite is green, the first question is whether the tests still test anything — §4 is the reason. Ask for the two lists (*what changed, what you verified*), and read the second.

Review by risk, not by line count. A 400-line change that adds a module is usually safer than a 4-line change to a boundary condition in shared code. Ask what a future session will be missing — the answer is a list of the things that lived in this conversation and nowhere else, and it is also your commit message.

Diagnose before editing. When something breaks: read the failure, state the likely cause, *then* fix. Jump straight to the fix and you get a repair that never names the assumption that broke, so the same bug comes back in a different shape.

Then decide whether the failure earns a guardrail. Not every bug does. A test, a rule in `CLAUDE.md`, a deny rule, a hook — each has an ongoing cost, and a guardrail added in irritation is paid for by everyone forever. The question is whether this failure will recur, not whether it was annoying.

The lab is deliberately arranged so that a review can *miss* the defect. That is data, not a bug in the exercise. A review that names the layer a defect lives in is useful even when it does not name the line.

12. Landing in a codebase you did not write

Lab 12. `legacy-service`: fifteen modules, five years of history, four tests, and several things that are not what they look like.

Every other lab happens in an app you built four minutes earlier. This is the honest one, and it is the most common thing you will do with an agentic tool in your first month.

Ask for a map, read-only. Four requirements, and the fourth is the one people leave out:

1. what it does,
2. **where execution starts** — the single most useful fact about an unfamiliar codebase, and the one a README most often has wrong,
3. which files to read first, so you get a *reading order* rather than a summary,
4. **where it is guessing versus where the code is explicit.**

In your own project you would notice a confident wrong answer. In a foreign repository you have no such reflex, so make it mark the seam. (Ch. 4 has his orientation prompt; the lab asks you to write yours from these four.)

Then mine the history, which the map will not do for you:

```
Look through the git history for <module> and summarise how its API came to be.  
What decisions does the history explain that the code does not?
```

Scoped to one module, because "summarise the history" of a five-year repo gets you a shrug. *How its API came to be* asks for a sequence rather than a summary. And *what the history explains that the code does not* is the point — you are mining for intent that never made it into a comment. A five-line commit message from 2021 can stop you "fixing" something deliberate.

Spot-check the map against the territory. If it says the request flow runs one way, open one file and follow it. If it calls something dead code, `grep` for the symbol before you believe it. In the lab's fixture, one module's name has been a lie since 2022, one "dead" module still has a live import from a cron script, and a comment asserts a rewrite the git history does not contain.

Then write down only what you verified, deleting the merely plausible. That file is the artifact — orientation is expensive, and paying for it twice is pure waste.

13. When not to use it

Lab 13. Write your stop list, and pick one metric.

The honest boundaries, from the material rather than from caution:

- **Where you cannot check the result.** The tool's failure mode is confident and plausible. If you are not equipped to tell a right answer from a convincing one — an unfamiliar domain, a language you do not read — you are not reviewing, you are agreeing.
- **Where the cost of being wrong is asymmetric.** Migrations, money, deletions, anything outward-facing. Not because it will get it wrong, but because you will not catch it in time when it does.
- **Where the work is the thinking.** If the hard part is deciding what should exist, generating three plausible versions of the wrong thing is not progress.
- **Where the context does not fit.** A change that depends on knowledge spread across a system nobody has written down is a research task first. Do §12, then come back.

The failure mode that actually costs teams is none of those. It is **decision fatigue**: fifty small approvals in an afternoon, each individually reasonable, and by the fortieth you are approving without reading. The defences are structural, not moral — permissions so the boring approvals never reach you, sessions short enough that you are still paying attention at the end, and a habit of reading the diff for the changes you did not ask for rather than the ones you did.

And pick one metric you will actually watch. Not lines generated. Something like: *how often does a change come back in review for something the tests should have caught?* If that number is going up, the speed is not real.

A closing note on the trade

This lab set is deliberately incomplete in one specific way: where a prompt is Kousen's, it is described rather than reproduced, and you write your own from the requirements. That is worse for you in the short run and better in the long run, because the skill being taught is not *having* the prompt. It is knowing which four things the prompt has to do.

If you own the book, read it alongside these. If you do not, the requirements in each brief are enough to write something that works.